

积分

概念 若 $F'(x) = f(x)$, 则 $f(x)$ 的全体原函数 $F(x) + C$ 为 $f(x)$ 的不定积分 $\int f(x) dx = F(x) + C$ *

$F(x)$: 原函数 $f(x)$: 被积函数 $\int f(x) dx$: 被积表达式 x : 积分变量 C : 积分常数 $\int$: 不定积分号 $F(x) + C$: 积分曲线族

几何意义: $f(x)$ 不定积分图形一族“平行”的积分曲线, 每个横坐标相同点处切线平行

定理: ① 若 $f(x)$ 在 I 上连续, 则 $f(x)$ 在 I 上可积

② 若 $F(x)$ 与 $G(x)$ 为 $f(x)$ 2 个原函数 $\exists C$, $F(x) = G(x) + C$

基本积分:

$\int x^\alpha dx = \frac{1}{\alpha+1} x^{\alpha+1} + C \quad (\alpha \neq -1)$	$\int \sec^2 x dx = \tan x + C$	$\int \sec x \tan x dx = \sec x + C$	$\int \sinh x dx = \cosh x + C$	$\int \sec x dx = \ln \sec x + \tan x + C$
$\int \frac{1}{x} dx = \ln x + C$	$\int \csc^2 x dx = -\cot x + C$	$\int \csc x \cot x dx = -\csc x + C$	$\int \cosh x dx = \sinh x + C$	$\int \csc x dx = \ln \csc x - \cot x + C$
$\int a^x dx = \frac{a^x}{\ln a} + C \quad (a > 0, a \neq 1)$				

性质: ① $d \int f(x) dx = f(x)$

② $\int d f(x) = f(x) + C$

③ 线性运算法则: $\int [x f(x) + y g(x)] dx = x \int f(x) dx + y \int g(x) dx$

求导检验!!!

思路: $\int f(u) g(u) du$

① 求 $f(u)$ 看与 $g(u)$ 关系
② 求 $\varphi(u)$

① 凑微分 (第一换元) 设 $F(u)$ 可微 $F(u) = f(u)$ $u = \varphi(x)$ 可微

$$\int f(\varphi(x)) \varphi'(x) dx = \int f(u) d(\varphi(x)) \stackrel{u=\varphi(x)}{=} \int f(u) du = F(u) + C = F(\varphi(x)) + C$$

常用微分关系式:

$$d(x+b) = dx$$

$$d(x^2) = 2x dx$$

$$d(\frac{1}{x}) = -\frac{1}{x^2} dx$$

$$d(\ln x) = \frac{1}{x} dx$$

$$d(\arcsin x) = \frac{1}{\sqrt{1-x^2}} dx$$

$$d(\arctan x) = \frac{1}{1+x^2} dx$$

$$d(\sec x) = \sec x \tan x dx$$

$$d(\csc x) = -\csc x \cot x dx$$

$$d(\sinh x) = \cosh x dx$$

$$d(\cosh x) = \sinh x dx$$

$$\cos x dx = d \sin x$$

$$\sin x dx = -d \cos x$$

$$\frac{1}{\sqrt{1-x^2}} dx = d \arcsin x$$

$$\frac{1}{1+x^2} dx = d \arctan x$$

$$\sec x dx = d \tan x$$

$$\csc x dx = d \cot x$$

$$\sinh x dx = d \cosh x$$

$$\cosh x dx = d \sinh x$$

$$\frac{1}{x} dx = d \ln x$$

$$e^x dx = d e^x$$

$$\frac{1}{\sqrt{1-x^2}} dx = d \arcsin x$$

$$\frac{1}{1+x^2} dx = d \arctan x$$

$$\sec x dx = d \tan x$$

$$\csc x dx = d \cot x$$

$$\sinh x dx = d \cosh x$$

$$\cosh x dx = d \sinh x$$

$$\frac{x}{1-x^2} dx = -\frac{1}{2} d \ln |1-x^2|$$

$$\frac{x}{1+x^2} dx = \frac{1}{2} d \ln |1+x^2|$$

$$\frac{1}{a^2-x^2} dx = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C$$

$$\frac{1}{a^2+x^2} dx = \frac{1}{a} \arctan \frac{x}{a} + C$$

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基本方法

② 换元法 (第二换元)

若 $x = \varphi(t)$ 严格单调可微且 $F(u) = f(\varphi(u)) \varphi'(u)$ 则 $\int f(x) dx = F(\varphi^{-1}(x)) + C$

① 1次 $\int \frac{dx}{1+x^2} = \int \frac{1}{1+t^2} dt = \int \frac{1}{1+t^2} dt = \arctan t + C = \arctan x + C$

② 2次 $\int \sqrt{a^2-x^2} dx = \int \sqrt{a^2-a^2 \sin^2 t} a \cos t dt = a^2 \int \cos^2 t dt = \frac{a^2}{2} \int (1 + \cos 2t) dt = \frac{a^2}{2} (t + \frac{1}{2} \sin 2t) + C = \frac{a^2}{2} (t + \sin t \cos t) + C = \frac{a^2}{2} (\arcsin \frac{x}{a} + \frac{x}{a} \sqrt{1-\frac{x^2}{a^2}}) + C$

$\int \frac{1}{\sqrt{x^2-a^2}} dx = \int \frac{1}{a \tan t} a \sec^2 t dt = \int \sec t dt = \ln |\sec t + \tan t| + C = \ln \left| \frac{x}{a} + \sqrt{\frac{x^2}{a^2} - 1} \right| + C$

$\int \frac{1}{(x^2+a^2)^{\frac{3}{2}}} dx = \int \frac{1}{a^3 \sec^3 t} a \sec^2 t dt = \frac{1}{a^2} \int \sec t dt = \frac{1}{a^2} \ln |\sec t + \tan t| + C = \frac{1}{a^2} \ln \left| \frac{x}{a} + \sqrt{\frac{x^2}{a^2} + 1} \right| + C$

幂 $\int x^2 (3x+5)^{100} dx = \int \frac{1}{3} (3t+5)^{100} dt = \frac{1}{3} \int (3t+5)^{100} dt = \frac{1}{3} \frac{(3t+5)^{101}}{101} + C = \frac{1}{303} (3x+5)^{101} + C$

化简 $\int (t^2+1)^{101} dt = \int (t^2+1)^{100} (t^2+1) dt = \int (t^2+1)^{100} t^2 dt + \int (t^2+1)^{100} dt = \frac{1}{103} (t^2+1)^{103} + \frac{2t}{101} (t^2+1)^{101} + C$

倒变换 $\int \frac{1}{x \sqrt{ax^2+bx+c}} dx$ 令 $x = \frac{1}{t}$

$\int \frac{1}{x \sqrt{ax^2+bx+c}} dx = \int \frac{1}{\frac{1}{t} \sqrt{a + bt + ct^2}} \cdot \frac{1}{t^2} dt = - \int \frac{1}{\sqrt{a + bt + ct^2}} dt = - \int \frac{1}{\sqrt{1 - (t+\frac{b}{2c})^2}} dt = - \arcsin \frac{t+\frac{b}{2c}}{1} + C$

$\int \frac{dx}{x \sqrt{x^2-1}} = \int \frac{1}{\frac{1}{t} \sqrt{1-t^2}} \cdot \frac{1}{t^2} dt = - \int \frac{1}{\sqrt{1-t^2}} dt = - \arcsin t + C = - \arcsin \frac{1}{x} + C$

思路:

① 被积函数含 $\sqrt{ax+b}$ 令 $t = \sqrt{ax+b} / ax+b$

② $\frac{1}{a^2-x^2} = \frac{1}{(a-x)(a+x)} = \frac{1}{2a} \left(\frac{1}{a-x} + \frac{1}{a+x} \right)$

$\frac{1}{x^2-a^2} = \frac{1}{(x-a)(x+a)} = \frac{1}{2a} \left(\frac{1}{x-a} - \frac{1}{x+a} \right)$

$\frac{1}{a^2+x^2} = \frac{1}{H(\frac{x}{a})^2 + 1} \cdot \frac{1}{a^2} = \frac{1}{a^2} \arctan \frac{x}{a}$

$\sqrt{a^2-x^2} = \frac{a \cos t}{\sin t} \quad x = a \sin t / a \cos t$

$\sqrt{a^2+x^2} = a \sec t \quad x = a \tan t$

$\sqrt{x^2-a^2} = a \tan t \quad x = a \sec t$

若分子有 x 幂次, 可直接整体换

②分部积分法 $\int u dv = uv - \int v du$

求导不变 求导化简

被积函数为两种不同类型函数积

① $P(x)$ 为多项式函数 $\int P(x) e^{ax} / \sin / \cos x dx$ 令 $u = P(x)$

② $P(x)$ 为多... $\int P(x) (\ln x)^n / \arctan x / \arcsin x dx$ 令 $dv = P(x) dx$

③ 连续几次分部积分后出现原式, 移项合并

$$\int x^3 \ln x dx = \frac{1}{4} \int \ln x dx^4 = \frac{1}{4} x^4 \ln x - \frac{1}{4} \int x^4 \frac{1}{x} dx = \frac{1}{4} x^4 (\ln x - \frac{1}{4}) + C$$

$$\int x e^x dx = \int x dx e^x = x e^x - \int e^x dx = x e^x - e^x + C$$

$$\int x \arctan x dx = \frac{1}{2} \int \arctan x d(x^2) = \frac{1}{2} \int x^2 d(\arctan x) = \frac{1}{2} x^2 \arctan x - \frac{1}{2} \int \frac{1}{1+x^2} dx = \frac{1}{2} x^2 \arctan x - \frac{1}{2} x + C$$

$$\int \arcsin x dx = x \arcsin x - \int x d(\arcsin x) = x \arcsin x - \int \frac{x}{\sqrt{1-x^2}} dx = x \arcsin x + \sqrt{1-x^2} + C$$

$$\int x^2 \sin x dx = -\int x^2 d \cos x = -x^2 \cos x + \int 2x \cos x dx = (x^2 \cos x - 2x \sin x + 2 \cos x) + C$$

$$\int e^x \sin x dx = \int \sin x d e^x = e^x \sin x - \int e^x \cos x dx = e^x \sin x - \int \cos x d e^x = e^x (\sin x - \cos x) - \int e^x \sin x dx$$

$$\therefore 2 \int e^x \sin x dx = e^x (\sin x - \cos x) + C_1$$

$$= \frac{e^x}{2} (\sin x - \cos x) + C$$

$$\int \sec^3 x dx = \int \sec x d \tan x = \sec x \tan x - \int \tan x (\sec^2 x - 1) dx = \sec x \tan x - \int \sec^3 x dx + \int \tan x dx$$

$$\therefore \int \sec^3 x dx = \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x| + C$$

$$\int \sqrt{a^2 - x^2} dx = x \sqrt{a^2 - x^2} - \int x \frac{-2x}{2\sqrt{a^2 - x^2}} dx$$

$$\int \frac{x \ln(x + \sqrt{1+x^2})}{\sqrt{1+x^2}} dx = \int \frac{\ln(x + \sqrt{1+x^2})}{2\sqrt{1+x^2}} d(1+x^2) = \int \ln(x + \sqrt{1+x^2}) d\sqrt{1+x^2} = \sqrt{1+x^2} \ln(x + \sqrt{1+x^2}) - \int \sqrt{1+x^2} \frac{1}{x + \sqrt{1+x^2}} dx = \sqrt{1+x^2} \ln(x + \sqrt{1+x^2}) - x$$

递推 $I_n = \int \frac{1}{(x^2+a^2)^n} dx$

当 $n=1$ 时 $I_1 = \int \frac{1}{x^2+a^2} dx = \frac{1}{a} \arctan \frac{x}{a} + C$

当 $n>1$ 时 $I_n = \int \frac{1}{(x^2+a^2)^n} dx = \frac{x}{(x^2+a^2)^{n-1} + 2(n-1)} \int \frac{(x^2+a^2)^{n-1} - a^2}{(x^2+a^2)^n} dx$

$$= \frac{x}{(x^2+a^2)^{n-1} + 2(n-1)} \left[\int \frac{1}{(x^2+a^2)^{n-1}} dx - \int \frac{a^2}{(x^2+a^2)^n} dx \right]$$

$$= \frac{x}{(x^2+a^2)^{n-1} + 2(n-1)} [I_{n-1} - a^2 I_n]$$

$$\therefore I_n = \frac{1}{2a^2(n-1)} \left[\frac{x}{(x^2+a^2)^{n-1}} + (n-3) I_{n-1} \right]$$

1. 真分式和假分式

$$\frac{P(x)}{Q(x)} = \frac{a_n x^n + \dots + a_1 x + a_0}{b_m x^m + \dots + b_1 x + b_0} \quad \begin{matrix} n < m & \text{真} \\ n \geq m & \text{假} \end{matrix}$$

2. 两类最简分式

① $\frac{A}{(x-a)^k}$ ② $\frac{Ax+B}{(x^2+px+q)^k}$ $P^2-4q < 0$ A, B, k 系数

3. 真分式 $\rightarrow$ 最简分式

把 $\frac{x^4+2x^3+2x+1}{x^2+x+1}$ 化为多项式与有理真分式之和

长除法 $\rightarrow x^2+x+2 + \frac{3x+3}{x^2+x+1}$ 只考虑有理真分式积分

把有理真分式 $\frac{x^2-5}{(x+1)(x+2)}$ 分解为最简分式之和

待定系数法 $\frac{x^2-5}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$

$$x^2-5 = (A+B)x^2 + (A+2B)x + (A+2B+C)$$

$$\begin{cases} A+B=1 \\ -4A+B+C=0 \\ A+2B+C=5 \end{cases} \therefore A=-\frac{4}{3}, B=\frac{7}{3}, C=-\frac{5}{3}$$

赋值法 $x^2-5 = A(x+2)^2 + B(x+1)(x+2) + C(x+1)$

$$\begin{cases} \text{令 } x=-2 & -1 = C(-1+1) & C=-\frac{1}{2} \\ \text{令 } x=-1 & -4 = A(-1+2)^2 & A=-4 \\ \text{令 } x=0 & -5 = 4A+2B+C & B=\frac{7}{3} \end{cases}$$

推广: $\frac{x^2-5}{(x+1)(x+2)(x+3)(x^2+4x+5)^2} = \frac{A}{x+1} + \frac{B}{x+2} + \frac{C}{x+3} + \frac{Dx+E}{x^2+4x+5} + \frac{Fx+G}{x^2+4x+5} + \frac{Hx+I}{(x^2+4x+5)^2} + \frac{Jx+K}{(x^2+4x+5)^2}$

1. 当 $n=1$ 时 $\int \frac{1}{x-a} dx = \ln|x-a| + C$

当 $n>1$ 时 $\int \frac{1}{(x-a)^n} dx = \int (x-a)^{-n} d(x-a) = \frac{1}{-n+1} (x-a)^{-n+1} + C$

2. 当 $n=1$ 时 $\int \frac{x+A}{x^2+px+q} dx$ ($P^2-4q < 0$)

当 $n>1$ 时 ($P^2-4q < 0$)

$$\int \frac{x+A}{x^2+px+q} dx = \frac{1}{2} \int \frac{2x+p}{x^2+px+q} dx + (A-\frac{p}{2}) \int \frac{1}{x^2+px+q} dx = \frac{1}{2} \int \frac{d(x^2+px+q)}{x^2+px+q} + (A-\frac{p}{2}) \int \frac{1}{(x+\frac{p}{2})^2 + (\frac{4q-p^2}{4})} d(x+\frac{p}{2})$$

$$= \frac{1}{2} \ln|x^2+px+q| + \frac{A-\frac{p}{2}}{\sqrt{\frac{4q-p^2}{4}}} \arctan \frac{x+\frac{p}{2}}{\sqrt{\frac{4q-p^2}{4}}} + C$$

定理: 一切有理函数的原函数都可以用多项式/有理/对数/反正切函数, 一定是初等函数

注: 初等函数定义域内原函数, 但原函数不一定是初等函数

例: $\int \frac{1}{e^{x+3}e^{x+2}} dx = \int \frac{1}{e^{2x+5}} dx = \int \frac{1}{e^{2x}e^5} dx = \frac{1}{e^5} \int \frac{1}{e^{2x}} dx = -\frac{1}{2e^5} e^{-2x} + C = -\frac{1}{2e^{2x+5}} + C$

例: $\int \frac{x^4}{(1+x^2)^2} dx = \frac{1}{2} \int \frac{x^2}{(1+x^2)^2} d(1+x^2) = -\frac{1}{2} \int \frac{x^2}{(1+x^2)^2} d(1+x^2) = -\frac{1}{2} \int \frac{1}{(1+x^2)^2} d(1+x^2) = -\frac{1}{2} \int \frac{1}{u^2} du = \frac{1}{2u} + C = \frac{1}{2(1+x^2)} + C$

下求 $\int \frac{1}{(1+x^2)^2} dx = \int \left(\frac{1}{1+x^2} - \frac{x}{(1+x^2)^2} \right) dx$

$$\int \frac{1}{1+x^2} dx = \frac{x}{1+x^2} - \int x \cdot \frac{-2x}{(1+x^2)^2} dx = \frac{x}{1+x^2} + 2 \int \frac{x^2}{(1+x^2)^2} dx = \frac{x}{1+x^2} + 2 \left(\frac{1}{2} \int \frac{1}{(1+x^2)^2} dx \right)$$

$$\therefore \int \frac{1}{(1+x^2)^2} dx = \frac{x}{2(1+x^2)} + \frac{1}{2} \arctan x + C$$

$$\int \frac{x \arctan x}{\sqrt{1+x^2}} dx = \int \arctan x d\sqrt{1+x^2}$$

$$= \sqrt{1+x^2} \arctan x - \int \frac{1}{\sqrt{1+x^2}} dx$$

$$= \sqrt{1+x^2} \arctan x - \frac{x}{\sqrt{1+x^2}} + C$$

$$\int x \ln(4+x^4) dx = \frac{1}{2} \int \ln(4+x^4) d(x^2)$$

$$= \frac{1}{2} x^2 \ln(4+x^4) - \frac{1}{2} \int x^2 \frac{4x^3}{4+x^4} dx$$

$$= \frac{1}{2} x^2 \ln(4+x^4) - 2 \int \left(x - \frac{4x}{4+x^4} \right) dx$$

$$= \frac{1}{2} x^2 \ln(4+x^4) - x^2 + 2 \arctan \frac{x}{\sqrt{2}} + C$$

分式

有理函数

有理函数积分

三角函数有理式

$$\int \frac{1}{\sqrt{(x-1)(x+1)^3}} dx = \int \frac{1}{(x-1)^{1/2} (x+1)^{3/2}} dx$$

$$\text{令 } t = \sqrt{\frac{x-1}{x+1}} \text{ 则 } x = \frac{t^2+1}{1-t^2} = \frac{2}{1-t^2} - 1 \quad dx = \frac{4t}{(1-t^2)^2} dt$$

$$\text{原式} = \int \frac{1}{(\frac{t^2+1}{1-t^2}-1)^{1/2} \cdot \frac{2}{(1-t^2)^{3/2}}} \cdot \frac{4t}{(1-t^2)^2} dt = \int \frac{2}{t} dt$$

① 万能公式 令 $t = \tan \frac{x}{2}$ $x = 2 \arctan t$ $\int R(\sin x, \cos x) dx = \int R(\frac{2t}{1+t^2}, \frac{1-t^2}{1+t^2}, \frac{2}{1+t^2}) \cdot \frac{2}{1+t^2} dt$

$$\int \frac{\sin x}{1 + \sin x + \cos x} dx$$

$$\int \frac{\sin x \cos x}{1 + \cos^2 x} dx = -\frac{1}{2} \int \frac{1}{1 + \cos^2 x} d \cos^2 x = -\frac{1}{2} \arctan(\cos^2 x) + C$$

② 特殊类

1. $\int \sin^m x \cos^n x dx$ 其中 m, n 至少有一个为奇数 另一个为任意实数

m 奇: $\int \sin^m x \cos^n x dx = -\int \sin^{m-1} x \cos^n x d \cos x = -\int (1 - \cos^2 x)^{\frac{m-1}{2}} \cos^n x d \cos x$

n 奇: $\int \sin^m x \cos^n x dx = \int \sin^m x \cos^{n-1} x d \sin x = \int \sin^m x (1 - \sin^2 x)^{\frac{n-1}{2}} d \sin x$

$$\int \sin^{\frac{1}{2}} x \cos^3 x dx = \int \sin^{\frac{1}{2}} x (1 - \sin^2 x) d \sin x$$

2. $\int \sin^m x \cos^n x dx$ 其中 m, n 均为偶数

倍角降次: $\cos^2 x = \sin^2 x / \sin x \cos x$

$$\int \sin^2 x \cos^4 x dx = \int (\sin x \cos x)^2 \cos^2 x dx = \int \frac{1}{2} \sin^2 2x \cdot \frac{1 + \cos 2x}{2} dx = \frac{1}{8} \int \sin^2 2x dx + \frac{1}{8} \int \sin^2 2x \cos 2x dx$$

3. $\int \sin^m x \cos^n x dx / \int \sin^m x \cos^n x / \cos^m x \sin^n x$ 其中 $m \neq n$ 积化和差

4. $\int R(\sin x, \cos x) dx$ 其中 $R(\sin x, \cos x) = R(t \sin x, t \cos x)$ 令 $t = \tan x$ 或同除 $\cos^n x$ 凑 $d(\tan x)$

$$R(\sin x, \cos x) = R(\tan x \cos x, \cos x) = R_1(\tan x, \cos x) = R_1(t \tan x, \cos x) = R_2(t \tan x, \cos^2 x) = R_3(t \tan x, \frac{1}{1 + \tan^2 x}) = R_4(t \tan x)$$

令 $t = \tan x$ $\int R(\sin x, \cos x) dx = \int R_4(t \tan x) dx$ $R_4(x, y)$ 中 y 仅以平方项出现

$$\int \frac{1}{2 - \cos^2 x} dx \xrightarrow{t = \tan x} \int \frac{1}{2 - \frac{1}{1+t^2}} \cdot \frac{1}{1+t^2} dt = \int \frac{1}{1+t^2} d(\frac{1}{2}) = \frac{1}{2} \arctan(\sqrt{2}t) + C$$

1. $\int R(x, \sqrt[n]{\frac{ax+b}{cx+d}}) dx$ 令 $\sqrt[n]{\frac{ax+b}{cx+d}} = t \Rightarrow x = \frac{dt^n - b}{a - ct^n}$

$$\int \frac{dx}{\sqrt[n]{(x-a)^{n+1} (x-b)^{n+1}}} = \int \frac{1}{(x-a)^{n+1} (x-b)^{n+1}} dx = \int \frac{1}{(x-a)(x-b)} \sqrt[n]{\frac{x-b}{x-a}} dx \quad t = \sqrt[n]{\frac{x-b}{x-a}}$$

2. $\int R(x, \sqrt{ax^2+bx+c}) dx$ 化为 $\sqrt{4b^2 \pm k^2}$ k 为 x 一次多项式 三角换元

① $a > 0$ 令 $\sqrt{ax^2+bx+c} = t + \sqrt{ax}$ 或 $t - \sqrt{ax}$

② $c > 0$ 令 $\sqrt{ax^2+bx+c} = x \sqrt{a} \sqrt{1 + \frac{b}{a}x + \frac{c}{a}}$ 或 $x \sqrt{a} \sqrt{1 - \frac{b}{a}x + \frac{c}{a}}$

③ 若 $\sqrt{ax^2+bx+c} = \sqrt{a(x-d)(x-f)}$ $\sqrt{a(x-d)(x-f)} = t(x+d)$ 或 $t(x-f)$

$$\int \frac{dx}{x + \sqrt{x^2 + x + 1}} \quad \text{令 } \sqrt{x^2 + x + 1} = t - x \Rightarrow x = \frac{t^2 - 1}{t + 2}$$

$$= \frac{1}{2} \int \frac{t^2 + 1}{t(t + \frac{1}{2})^2} dt = \frac{1}{2} \int [\frac{t}{t} - \frac{3}{t + \frac{1}{2}} - \frac{2}{2(t + \frac{1}{2})^2}] dt = \frac{1}{2} \ln |t + \frac{1}{2}| + \frac{3}{2(t + \frac{1}{2})} + C$$

分段求 $\int \max\{1, x^2\} dx$

$$f(x) = \begin{cases} x^2 & x < -1 \\ 1 & -1 \leq x \leq 1 \\ x^2 & x > 1 \end{cases} \Rightarrow f(x) = \begin{cases} \frac{1}{3}x^3 + C_1 & x < -1 \\ x + C_2 & -1 \leq x \leq 1 \\ \frac{1}{3}x^3 + C_3 & x > 1 \end{cases}$$

连续: $-\frac{1}{3} + C_1 = -1 + C_2 = f(-1)$
 $\frac{1}{3} + C_2 = 1 + C_3 = f(1)$

$f(x) = \begin{cases} 1 & 0 < x < 1 \\ x & x > 1 \end{cases}$ 且 $f(0) = 0$, 求 $f(x)$

令 $t = f(x)$

$$f(t) = \begin{cases} 1 & 0 < t < 1 \\ e^t & t > 1 \end{cases} = \begin{cases} 1 & -\infty < t < 0 \\ e^t & 0 < t < \infty \end{cases} \Rightarrow f(t) = \begin{cases} t + C_1 & -\infty < t < 0 \\ e^t + C_2 & 0 < t < \infty \end{cases}$$

连续: $C_1 = 1, C_2 = 0$

抽象 $\int x f'(x) dx$

$$= \int x d f(x) = x f(x) - \int f(x) dx = x f(x) - f(x) + C$$

无理函数

其他

① 在 $[a, b]$ 内任插 $n-1$ 个分点

任给 $\xi_i \in [x_{i-1}, x_i]$ $\Delta x_i = x_i - x_{i-1}$ 积分元 $f(\xi_i) \Delta x_i$

积分和式 $f(\xi_1) \Delta x_1 + \dots + f(\xi_n) \Delta x_n = \sum_{i=1}^n f(\xi_i) \Delta x_i$

设 $\lambda = \max\{\Delta x_i : 1 \leq i \leq n\}$ $I = \int_a^b f(x) dx = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(\xi_i) \Delta x_i$ 注: 不能保证 $\lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i) \Delta x_i = S$

若 $\lambda \rightarrow 0$ 时, I 与 $[a, b]$ 分法 ξ_i 取法无关

则称 f 在 $[a, b]$ 上可积

② 黎曼积分

① 常数 $I, \forall \epsilon > 0, \exists \delta > 0, \forall$ 分割 $T: a = x_0 < x_1 < \dots < x_{n-1} < x_n = b$

只要 $\max\{\Delta x_i : 1 \leq i \leq n\} < \delta$ 就有 $| \sum_{i=1}^n f(\xi_i) \Delta x_i - I | < \epsilon$

五、定积分

定义

则称 I 为 $f(x)$ 在 $[a, b]$ 上的定积分, 记作 $\int_a^b f(x) dx$, 与 $f(x)$ 和 $[a, b]$ 有关, 与积分法无关

规定 若 $a > b$ $\int_a^b f(x) dx = -\int_b^a f(x) dx$ $\int_a^a f(x) dx = 0$

⑤ 几何意义: 图像面积

$f(x) \geq 0$ 面积

$f(x) \leq 0$ 负面积

时 ≥ 0 时 ≤ 0 代数和

④ 实际应用:

曲边梯形 S / 变力做功 / 变速直线运动 S

必要: 有界

条件 < 充分: 连续 / 单调 / 间断点个数有限 + 有界

可积函数类

用定义算

$$\int_0^1 e^x dx$$

将 $[0, 1]$ n 等分, 分点 $x_i = \frac{i}{n}$ ($i=1, 2, \dots, n$) $\Delta x_i = \frac{1}{n}$ 取右端点 $\xi_i = \frac{i}{n}$

$$\begin{aligned} \int_0^1 e^x dx &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i) \Delta x_i = \lim_{n \rightarrow \infty} \sum_{i=1}^n e^{\frac{i}{n}} \cdot \frac{1}{n} \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} (e^{\frac{1}{n}} + e^{\frac{2}{n}} + \dots + e^{\frac{n}{n}}) = \lim_{n \rightarrow \infty} \frac{1}{n} \frac{e^{\frac{1}{n}}(1 - e^{\frac{n}{n}})}{(1 - e^{\frac{1}{n}})} = e - 1 \end{aligned}$$

极限表达式为 n 项和 (乘积取对数)

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{1^p + 2^p + \dots + n^p}{n^{p+1}} \quad (p > 0) &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i) \Delta x_i = \int_0^1 x^p dx \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n (\frac{i}{n})^p \cdot \frac{1}{n} = \int_0^1 x^p dx \end{aligned}$$

$$\textcircled{1} \int_a^b 1 dx = \int_a^b dx = b - a$$

② 设 $f(x), g(x)$ 在 $[a, b]$ 上可积, $\forall \alpha, \beta$, 则 $\alpha f(x) + \beta g(x)$ 在 $[a, b]$ 上可积, 且 $\int_a^b (\alpha f(x) + \beta g(x)) dx = \alpha \int_a^b f(x) dx + \beta \int_a^b g(x) dx$ 线性运算法则

设 $F(x) = \alpha f(x) + \beta g(x)$

$$\begin{aligned} \lim_{n \rightarrow \infty} \sum_{i=1}^n F(\xi_i) \Delta x_i &= \lim_{n \rightarrow \infty} \sum_{i=1}^n [\alpha f(\xi_i) + \beta g(\xi_i)] \Delta x_i \\ &= \lim_{n \rightarrow \infty} [\alpha \sum_{i=1}^n f(\xi_i) \Delta x_i + \beta \sum_{i=1}^n g(\xi_i) \Delta x_i] \\ &= \alpha \int_a^b f(x) dx + \beta \int_a^b g(x) dx \end{aligned}$$

推论: $\int_a^b (f(x) \pm g(x)) dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$ $\int_a^b \alpha f(x) dx = \alpha \int_a^b f(x) dx$

③ $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$ 区间可加性

注: 右不一定可拆成左, 需 c 为分点才可拆, 但左和一定等于右

讨论 a, b, c 位置

$$\textcircled{1} a < c < b \quad \int_a^b f(x) dx = \lim_{n \rightarrow \infty} [\sum_{i=1}^k f(\xi_i) \Delta x_i + \sum_{i=k+1}^n f(\xi_i) \Delta x_i] = \sum_{i=1}^k f(\xi_i) \Delta x_i + \sum_{i=k+1}^n f(\xi_i) \Delta x_i = \int_a^c f(x) dx + \int_c^b f(x) dx$$

让 c 为固定分点

$$\textcircled{2} c < a < b \quad \int_c^b f(x) dx = \int_c^a f(x) dx + \int_a^b f(x) dx$$

$$-\int_a^c f(x) dx = \int_c^b f(x) dx - \int_a^b f(x) dx$$

其它情况类似

④ 若 $f(x)$ 在 $[a, b]$ 上可积, 且 $f(x) \geq 0$, 则 $\int_a^b f(x) dx \geq 0$ 可积 $+ \geq 0 \Rightarrow \geq 0$

$$\because f(\xi_i) \geq 0 \quad \Delta x_i \geq 0 \quad \therefore \sum_{i=1}^n f(\xi_i) \Delta x_i \geq 0$$

⑤ 若 $f(x), g(x)$ 在 $[a, b]$ 上可积, 且 $f(x) \geq g(x)$, 则 $\int_a^b f(x) dx \geq \int_a^b g(x) dx$ 可积 $+ \geq \Rightarrow \geq$

$$\textcircled{6} \textcircled{4} \quad \int_a^b f(x) dx - \int_a^b g(x) dx = \int_a^b (f(x) - g(x)) dx \geq 0$$

⑥ 若 $f(x)$ 在 $[a, b]$ 上连续, $f(a) > 0$ 但 $f(x)$ 不 > 0 , 则 $\int_a^b f(x) dx > 0$ 连续 $+ > 0 + \neq 0 \Rightarrow > 0$

$$\exists x_0 \in (a, b) \quad \lim_{x \rightarrow x_0} f(x) = f(x_0) > 0$$

由保号性 $\exists \delta > 0 \quad \exists \eta > 0$ 当 $x \in (x_0 - \delta, x_0 + \delta)$ 时, $f(x) > \eta > 0$

$$\begin{aligned} \int_a^b f(x) dx &= \int_a^{x_0-\delta} f(x) dx + \int_{x_0-\delta}^{x_0+\delta} f(x) dx + \int_{x_0+\delta}^b f(x) dx \\ &\geq \int_{x_0-\delta}^{x_0+\delta} f(x) dx \geq \int_{x_0-\delta}^{x_0+\delta} \eta dx = \eta \cdot 2\delta = \eta \cdot \Delta x > 0 \end{aligned}$$

推论: 若 $f(x), g(x)$ 在 $[a, b]$ 上连续, $f(x) \geq g(x)$ 且 $f(x) \neq g(x)$, 则 $\int_a^b f(x) dx > \int_a^b g(x) dx$ 不算比大小

⑦ 若 $f(x)$ 在 $[a, b]$ 上可积, 则 $|\int_a^b f(x) dx| \leq \int_a^b |f(x)| dx$ 里大外小

$$\textcircled{8} -|f(x)| \leq f(x) \leq |f(x)| \Rightarrow -\int_a^b |f(x)| dx \leq \int_a^b f(x) dx \leq \int_a^b |f(x)| dx$$

⑧ 若 $f(x)$ 在 $[a, b]$ 上可积, $m \leq f(x) \leq M, x \in [a, b]$, 则 $m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$

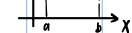
$$\textcircled{9} \int_a^b m dx \leq \int_a^b f(x) dx \leq \int_a^b M dx$$

⑨ 若 $f(x)$ 在闭区间 $[a, b]$ 上连续, 则至少存在一点 $\xi \in [a, b]$, 使 $\int_a^b f(x) dx = f(\xi)(b-a)$ 积分中值定理

$$\textcircled{10} m(b-a) \leq \int_a^b f(x) dx \leq M(b-a) \Rightarrow m \leq \frac{\int_a^b f(x) dx}{b-a} \leq M$$

至少一点 $\xi \in [a, b]$ $\frac{\int_a^b f(x) dx}{b-a} = f(\xi)$

几何意义:



$f(\xi)$ 为 $f(x)$ 在 $[a, b]$ 上的平均值

⑩ 若 $f(x)$ 在 $[0, 1]$ 上连续, 在 $(0, 1)$ 内可导, 且 $\int_0^1 f(x) dx = f(0)$, 证: $\exists \xi \in (0, 1)$ 使 $f'(\xi) = 0$

$$\because \exists c \in [0, 1], \text{使} \int_0^1 f(x) dx = f(c) \cdot \frac{1}{1-0} = f(c) \quad \therefore f(c) = f(0)$$

$$\therefore \exists \xi \in (0, c), f'(\xi) = 0$$

定义 2 设 $(\cdot, \cdot)$ 是欧氏空间 V 的一个内积运算, $\alpha \in V$, 称 $\sqrt{(\alpha, \alpha)}$ 为 α 的长度, 并记做 $|\alpha| = \sqrt{(\alpha, \alpha)}$

关于向量的内积与向量长度之间的关系, 我们有

定理 1 (Cauchy-Schwarz 不等式) 设 V 是定义了内积 $(\cdot, \cdot)$ 的欧氏空间, 则

$$|(\alpha, \beta)| \leq |\alpha| |\beta|, \quad \forall \alpha, \beta \in V. \quad (6)$$

(6) 取等号的充分必要条件是 α 与 β 线性相关.

证明 当 $\alpha = \beta = 0$ 时, (6) 显然成立; 当 α, β 不全为 0 时, 依内积的对称性质, 我们不妨设 $\alpha \neq 0$, 于是,

$$|\alpha|^2 t^2 + 2(\alpha, \beta)t + |\beta|^2 = (\beta + t\alpha, \beta + t\alpha) \geq 0, \quad \forall t \in \mathbb{R}. \quad (7)$$

这说明上述关于 t 的实系数二次多项式的判别式

$$\Delta = 4(\alpha, \beta)^2 - 4|\alpha|^2 |\beta|^2 \leq 0,$$

由此, (6) 当 α, β 不全为 0 时亦成立. (6) 得证.

当 (6) 取等号时, 若 $\alpha = \beta = 0$, 则 α, β 线性相关; 若 α, β 不全为 0, 同样依据内积的对称性, 不妨设 $\alpha \neq 0$, 依 (7) 知, 二次多项式 $(\beta + t\alpha, \beta + t\alpha) = 0$ 有重根 t_0 , 即

$$(\beta + t_0\alpha, \beta + t_0\alpha) = 0,$$

故 $\beta + t_0\alpha = 0$, 即 α, β 线性相关. 总之, 若 (6) 取等号, 则 α 与 β 线性相关. 必要性得证.

反之, 若 α 与 β 线性相关, 则存在不全为零的数 c_1, c_2 使得 $c_1\alpha + c_2\beta = 0$. 若 $c_1 \neq 0$, 则 $\alpha = -\frac{c_2}{c_1}\beta$, 于是

$$|(\alpha, \beta)| = |(\frac{c_2}{c_1}\beta, \beta)| = |\frac{c_2}{c_1}| |\beta|^2 = |\alpha| \cdot |\beta|,$$

即 (6) 取等号. 同理可证 (6) 当 $c_2 \neq 0$ 时亦取等号. 综上所述, 此时 (6) 取等号. 充分性得证. $\square$

补充: $[\int_a^b f(x)g(x) dx]^2 \leq \int_a^b f^2(x) dx \int_a^b g^2(x) dx$

欧氏空间中 Cauchy-Schwarz 不等式

$$x = (x_1, \dots, x_n) \in \mathbb{R}^n \quad y = (y_1, \dots, y_n) \in \mathbb{R}^n \quad (\sum_{i=1}^n x_i y_i)^2 \leq (\sum_{i=1}^n x_i^2) (\sum_{i=1}^n y_i^2)$$

$C[a, b]$ 中的 Cauchy-Schwarz 不等式

$$|\int_a^b f(x)g(x) dx| \leq (\int_a^b f^2(x) dx)^{\frac{1}{2}} (\int_a^b g^2(x) dx)^{\frac{1}{2}}$$

① 二次 $\Delta \leq 0 \quad \int_a^b (tf(x) + g(x))^2 dx \geq 0 \quad \Delta \geq 0$

$$\textcircled{2} \text{ 取 } |x_i y_i| \leq \sqrt{x_i^2 y_i^2} \quad (\geq x_i^2 y_i^2)$$

$$\int_a^b f^2(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f^2(\xi_i) \Delta x_i$$

$$\int_a^b g^2(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n g^2(\xi_i) \Delta x_i$$

$$\int_a^b f(x)g(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i)g(\xi_i) \Delta x_i$$

$\therefore \sum_{i=1}^n f^2(\xi_i) \Delta x_i \sum_{i=1}^n g^2(\xi_i) \Delta x_i \geq (\sum_{i=1}^n f(\xi_i)g(\xi_i) \Delta x_i)^2$ 两边 $\times \frac{1}{n}$ 再求极限

$$\textcircled{3} F(x) = \int_a^x f(x)g(x) dx = [\int_a^x f(x)g(x) dx]^2$$

$$F(x) = f(x) \int_a^x g(x) dx + g(x) \int_a^x f(x) dx - 2 \int_a^x f(x)g(x) dx$$

$$= \int_a^x (f(x)g(x) - g(x)f(x))^2 dx \geq 0, \quad \text{且} \quad F(x) \geq F(a) = 0$$

$\lim_{n \rightarrow \infty} \int_0^{2\pi} \frac{\sin nx}{x^2 + n^2} dx = 0$ Tip: $\sin x$ 无穷小时, 常用 $|\sin x| \leq |x|$
 $\leq \lim_{n \rightarrow \infty} \int_0^{2\pi} \frac{1}{x^2 + n^2} dx \leq \lim_{n \rightarrow \infty} \int_0^{2\pi} \frac{1}{n^2} dx = \lim_{n \rightarrow \infty} \frac{2\pi}{n^2} = 0$ 非无穷小
 $\geq \lim_{n \rightarrow \infty} \int_0^{2\pi} \frac{-1}{x^2 + n^2} dx \leq \lim_{n \rightarrow \infty} \frac{-2\pi}{n^2} = 0$

变上限的函数 $f(x)$ 在 I 上连续, $f(x)$ 在 $[a, x]$ 或 $[x, a]$ 上可积, $\forall x \in I$, 有唯一 $\int_a^x f(t) dt$ 与 x 对应 $\Rightarrow$ 函数 $G(x) = \int_a^x f(t) dt$

定理1 原函数存在定理 $f(x)$ 在区间 I 上连续, $a \in I$ 为一固定点, 则由变上限积分

$$G(x) = \int_a^x f(t) dt, x \in I \text{ 在 } I \text{ 上可导, 且 } G'(x) = f(x)$$

$$\lim_{\Delta x \rightarrow 0} \frac{G(x+\Delta x) - G(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\int_a^{x+\Delta x} f(t) dt - \int_a^x f(t) dt}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\int_x^{x+\Delta x} f(t) dt}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(\xi) \Delta x}{\Delta x} = \lim_{\Delta x \rightarrow 0} f(\xi) = f(x)$$

$$(\int_a^x f(t) dt)' = f(x)$$

推论: 原函数存在性定理: 连续函数一定存在原函数 $\int_a^x f(t) dt + C$

$$\text{变下限 } \frac{d}{dx} \int_x^b f(t) dt = \frac{d}{dx} [-\int_b^x f(t) dt] = -f(x)$$

$$\text{复合} = \frac{d}{dx} \int_a^{\alpha(x)} f(t) dt = f[\alpha(x)] \cdot \alpha'(x)$$

$$\frac{d}{dx} \int_{\alpha(x)}^{\beta(x)} f(t) dt = \frac{d}{dx} \left(\int_a^{\beta(x)} f(t) dt + \int_{\alpha(x)}^a f(t) dt \right) = f(\beta(x)) \beta'(x) - f(\alpha(x)) \alpha'(x)$$

$$\lim_{x \rightarrow 0} \frac{\int_0^x \frac{\sin t}{\sqrt{1+t^2}} dt}{\frac{x^2}{x^2}} = \lim_{x \rightarrow 0} \frac{\frac{\sin(x)}{\sqrt{1+x^2}} \cdot 2x}{\frac{2x}{4x^2}} = \lim_{x \rightarrow 0} \frac{1}{2\sqrt{1+x^2}} = \frac{1}{2}$$

定理2=牛顿莱布尼茨公式 $f(x)$ 在 $[a, b]$ 上连续且 $F(x)$ 为它在该区间原函数, 则 $\int_a^b f(x) dx = F(b) - F(a)$

$$\int_a^x f(x) dx - F(x) \equiv C$$

$$\text{令 } x=a \quad \int_a^a f(x) dx - F(a) \equiv C \Rightarrow C = -F(a) \Rightarrow \int_a^x f(x) dx = F(x) - F(a)$$

$$\text{令 } x=b \quad \int_a^b f(x) dx = F(b) - F(a)$$

$$\int_0^{\frac{\pi}{2}} (x^2 + 2 \cos x) dx = \left(\frac{1}{3} x^3 + 2 \sin x \right) \Big|_0^{\frac{\pi}{2}} = \frac{\pi^3}{24} + 2$$

$$\text{分段 } f(x) = \begin{cases} \sin x & 0 \leq x \leq \frac{\pi}{2} \\ x & \frac{\pi}{2} < x \leq \pi \end{cases} \quad \text{求 } \int_0^{\pi} f(x) dx = \int_0^{\frac{\pi}{2}} \sin x dx + \int_{\frac{\pi}{2}}^{\pi} x dx$$

$$\int_0^{\pi} f(x) dx = \int_0^{\frac{\pi}{2}} \sin x dx + \int_{\frac{\pi}{2}}^{\pi} x dx = 1 + \frac{1}{2} (\pi^2 - \frac{\pi^2}{4}) = 1 + \frac{3}{8} \pi^2$$

Tip 改变有限个点的函数值, 不改变定积分值

$$f(x) = g(x) \quad x \neq x_0 \quad x_0 \in [X_1, X_2]$$

$$\sum_{i=1}^n f(\xi_i) \Delta x_i = \begin{cases} \sum_{i=1}^n f(\xi_i) \Delta x_i & \xi_k \neq x_0 \\ \sum_{i=1}^n f(\xi_i) \Delta x_i - f(x_0) \Delta x_k + g(x_0) \Delta x_k & \xi_k = x_0 \end{cases}$$

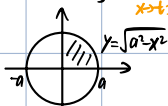
$$\int_a^b g(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n g(\xi_i) \Delta x_i = \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n f(\xi_i) \Delta x_i - f(x_0) \Delta x_k + g(x_0) \Delta x_k \right) = \int_a^b f(x) dx$$

①换元法 $x = \varphi(t)$ 满足 (1) $\varphi(a) = a, \varphi(b) = b$ (2) 在 $[a, b]$ 上有连续 $\varphi'(t)$

$$\text{则 } \int_a^b f(x) dx = \int_a^b f(\varphi(t)) \varphi'(t) dt = F(\varphi(t)) \Big|_a^b = F(b) - F(a) \text{ 不改变上下限}$$

$$\int_0^a \sqrt{a^2 - x^2} dx \xrightarrow{x = a \sin t} \int_0^{\frac{\pi}{2}} a \cos t \cdot a \cos t dt = a^2 \int_0^{\frac{\pi}{2}} \cos^2 t dt = \frac{a^2}{2} \left(t + \frac{\sin 2t}{2} \right) \Big|_0^{\frac{\pi}{2}} = \frac{\pi a^2}{4}$$

几何意义:



②分部积分 $\int_a^b u dv = uv \Big|_a^b - \int_a^b v du$

$$\int_0^{2\pi} x^2 \cos x dx = x^2 \sin x \Big|_0^{2\pi} - \int_0^{2\pi} \sin x \cdot 2x dx$$

$$\int_0^{\frac{\pi}{2}} \frac{x}{1 + \cos x} dx = \frac{1}{2} \int_0^{\frac{\pi}{2}} x \sin 2x dx$$

③二重积分

$$f(x) = \int_0^x \frac{\sin t}{\pi - t} dt, \text{ 求 } \int_0^{\pi} f(x) dx$$

$$\int_0^{\pi} f(x) dx = x f(x) \Big|_0^{\pi} - \int_0^{\pi} x f'(x) dx = \pi \int_0^{\pi} \frac{\sin t}{\pi - t} dt - \int_0^{\pi} \frac{x \sin x}{\pi - x} dx = \int_0^{\pi} \left(\frac{\pi \sin x}{\pi - x} - \frac{x \sin x}{\pi - x} \right) dx = \int_0^{\pi} \sin x dx = 2$$

不可积则定抵消

④简化方法

(1) 关于原点对称

$$\int_{-a}^a f(x) dx = \int_{-a}^a [f(x) + f(-x)] dx = \begin{cases} 0 & \text{奇} \\ 2 \int_0^a f(x) dx & \text{偶} \end{cases}$$

证 $\int_{-a}^a f(x) dx = 0$

$$\int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx = \int_0^a [f(-t) + f(t)] dt$$

分成积分区间长度相等两段

$$\int_0^a f(t) dt + \int_0^a f(t) dt = 2 \int_0^a f(t) dt$$

换元统一积分区间 (用已知条件)
将两段抵消为0

$$\int_{-2}^2 (|x| + x) e^{-|x|} dx = \int_{-2}^2 (|x| e^{-|x|} + x e^{-|x|}) dx = 2 \int_0^2 x e^{-x} dx + 0 = 2 \left(-x e^{-x} - e^{-x} \right) \Big|_0^2 = 2(1 - 3e^{-2})$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x}{1 + e^x} dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x}{1 + e^x} dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 x \left[\frac{1 + e^x + 1 + e^{-x}}{(1 + e^x)(1 + e^{-x})} \right] dx = \int_0^{\frac{\pi}{2}} \sin^2 x dx$$

(2) 周期函数

$$\forall a \in \mathbb{R}, \int_a^{a+T} f(x) dx = \int_0^T f(x) dx$$

Summary =

$$\textcircled{1} \int_{-a}^a f(x) dx = \int_{-a}^a [f(x) + f(-x)] dx = \begin{cases} 0 & \text{奇} \\ 2 \int_0^a f(x) dx & \text{偶} \end{cases}$$

$$\textcircled{2} \int_a^{a+T} f(x) dx = \int_0^T f(x) dx$$

$$\textcircled{3} \int_0^{\frac{\pi}{2}} \sin^n x dx = \int_0^{\frac{\pi}{2}} \cos^n x dx = \frac{n-1}{n} \cdot \frac{n-3}{n-2} \dots < \begin{cases} \frac{3}{2} \cdot 1 & \text{奇} \\ \frac{1}{2} \cdot \frac{3}{2} & \text{偶} \end{cases}$$

$$\int_a^{at} f(x) dx = \int_a^t f(x) dx + \int_t^{at} f(x) dx = \int_a^t f(x) dx + \int_t^{at} f(x) dx = \int_a^t f(x) dx$$

$$\int_0^{\frac{\pi}{2}} \sin^n x dx = \int_0^{\frac{\pi}{2}} \cos^n x dx = \frac{n-1}{n} \frac{n-3}{n-2} \dots$$

记 $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$

当 $n \geq 2$ $I_n = -\int_0^{\frac{\pi}{2}} \sin^{n-1} x d \cos x = -\sin^{n-1} x \cos x \Big|_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} (n-1) \sin^{n-2} x \cos x dx = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x dx$

$\Rightarrow I_n = \frac{n-1}{n} I_{n-2} \quad (n \geq 2)$ $= (n-1) \int_0^{\frac{\pi}{2}} (\sin^{n-2} x - \sin^n x) dx = (n-1) I_{n-2} - (n-1) I_n$

$\int_{-1}^1 x^4 \sqrt{1-x^2} dx$ $\int_{-1}^1 x^4 \sqrt{1-x^2} dx$

$= 2 \int_0^1 x^4 \sqrt{1-x^2} dx \xrightarrow{x=\sin t} 2 \int_0^{\frac{\pi}{2}} \sin^4 t \cos^2 t dt = 2 \int_0^{\frac{\pi}{2}} (\sin^2 t - \sin^4 t) dt = 2(I_2 - I_4)$

证 $\int_0^{2\pi} \sin^n x dx = 2 \int_0^{\pi} \sin^n x dx = 4 \int_0^{\frac{\pi}{2}} \sin^n x dx$

$= \int_{-\pi}^{\pi} \sin^n x dx = 2 \int_0^{\pi} \sin^n x dx = 2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^n x dx = 4 \int_0^{\frac{\pi}{2}} \sin^n x dx$

(4) 变量代换

设 $f(x)$ 在 $[0,1]$ 上连续, 证明 $\int_0^{\pi} x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx$

令 $t = \pi - x$ 换元

$\int_0^{\pi} x f(\sin x) dx = \int_{\pi}^0 (\pi - t) f(\sin(\pi - t)) (-1) dt$

$= \int_0^{\pi} (\pi f(\sin x) - x f(\sin x)) dx = \pi \int_0^{\pi} f(\sin x) dx - \int_0^{\pi} x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx$

$\int_0^{\frac{\pi}{2}} \frac{\ln(\tan x)}{1+\tan^2 x} dx$ $\xrightarrow{x=\tan t} \int_0^{\frac{\pi}{2}} \frac{\ln(\tan t)}{\sec^2 t} \sec^2 t dt$

$= \int_0^{\frac{\pi}{2}} \ln \frac{\cos t + \sin t}{\cos t} dt = \int_0^{\frac{\pi}{2}} \ln \frac{1 + \cos(\frac{\pi}{2}-t)}{\cos t} dt = \int_0^{\frac{\pi}{2}} \ln \cos t dt + \int_0^{\frac{\pi}{2}} \ln \cos(\frac{\pi}{2}-t) dt - \int_0^{\frac{\pi}{2}} \ln \cos t dt = \frac{\pi}{2} \ln 2$

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin t}{\sin t + \cos t} dt$

$\xrightarrow{t=\frac{\pi}{2}-u} \int_{\frac{\pi}{2}}^0 \frac{\sin(\frac{\pi}{2}-u)}{\sin(\frac{\pi}{2}-u) + \cos(\frac{\pi}{2}-u)} (-1) du = \int_0^{\frac{\pi}{2}} \frac{\cos u}{\cos u + \sin u} du = \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{\sin t}{\sin t + \cos t} dt + \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{\cos t}{\sin t + \cos t} dt = \frac{1}{2} \int_0^{\frac{\pi}{2}} dt = \frac{\pi}{4}$

定义 $Q = \int_a^b f(x) dx$

令 $Q(x) = \int_a^x f(t) dt$ $\Delta Q = Q(x+\Delta x) - Q(x) = \Delta Q + o(\Delta x)$ $\Delta x \rightarrow 0$

$dQ = f(x) dx = f(x) \Delta x$ 微分

微分法 ① $dQ = f(x) dx$ ② $Q = \int_a^b dQ = \int_a^b f(x) dx$

X型

$A = \int_a^b (f(x) - g(x)) dx = \int_a^b (y_2 - y_1) dx$

Y型

$A = \int_a^b (x_2 - x_1) dy$

算 $y^2=2x$ 与 $y=x-4$ 所围面积

① x 为积分变量 $\begin{cases} y^2=2x \\ y=x-4 \end{cases} \Rightarrow (2, -2), (8, 4)$

$A = \int_2^8 (\sqrt{2x} - (-\sqrt{2x})) dx + \int_2^8 (x-4) dx = 18$ $x_2 = \sqrt{2x}$ x_1 为分段函数

② y 为积分变量

$A = \int_{-2}^4 (4 + y - \frac{y^2}{2}) dy = 18$ $x_2 = \frac{1}{2}y^2$ $x_1 = y+4$

$A = \int_{-\frac{1}{2}}^1 (2 - \frac{1}{y}) dy + \int_1^2 (0 - y) dy = \frac{3}{2} \ln 2$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a>0, b>0$) 所围

① $A = 4 \int_0^a y dx = \frac{4b}{a} \int_0^a \sqrt{a^2 - x^2} dx = \frac{4b}{a} \cdot \frac{\pi a^2}{4} = \pi ab$

② $\begin{cases} x = a \cos t \\ y = a \sin t \end{cases} 0 \leq t \leq 2\pi$ $A = 4 \int_0^a y dx \xrightarrow{x=a \cos t} 4 \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} b \sin t \cdot a \cdot (-\sin t) dt$

Summary

平面面积 $S = \int_a^b (y_2 - y_1) dx = \int_a^b (x_2 - x_1) dy$

截面体 $V = \int_a^b A(x) dx$

弧微分 $ds = \sqrt{dx^2 + dy^2} = \sqrt{r^2 \dot{\theta}^2 + \dot{r}^2} dt = \sqrt{1 + \dot{\theta}^2} r dt = \sqrt{1 + \dot{\theta}^2} dy = \sqrt{r_0^2 + \dot{r}^2} dt$

弧长 $S = \int \sqrt{dx^2 + dy^2} = \int \sqrt{r^2 \dot{\theta}^2 + \dot{r}^2} dt = \int \sqrt{1 + \dot{\theta}^2} r dx = \int \sqrt{1 + \dot{\theta}^2} dy = \int \sqrt{r_0^2 + \dot{r}^2} dt$

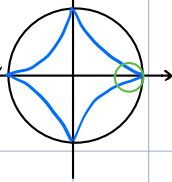
曲边扇形 $S = \frac{1}{2} r^2 \theta$

旋转体侧面积 $S = 2\pi \int_a^b y \sqrt{1+y'^2} dx$ 2017.5

旋转体体积 $V_x = \pi \int_a^b y^2 dx$ $V_y = 2\pi \int_a^b x f(x) dx$ [2017.5]

应用

参数方程 $\begin{cases} x = a \cos^2 t \\ y = a \sin^2 t \end{cases}$ 围成面积

$$A = 4 \int_0^{\frac{\pi}{2}} y dx = 4 \int_0^{\frac{\pi}{2}} a \cos^2 t \cdot (-2a \cos t \sin t) dt = -8a^2 \int_0^{\frac{\pi}{2}} \cos^3 t \sin t dt = 8a^2 \int_0^{\frac{\pi}{2}} \cos^2 t \sin t dt = \frac{8}{3} a^2$$


已知平行截面积求立体体积



$$V = \lim_{n \rightarrow \infty} \sum_{i=1}^n A(x_i) \Delta x_i = \int_a^b A(x) dx$$



$$x^2 + y^2 = a^2$$

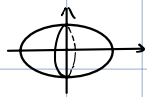
$$\rho a = \sqrt{a^2 - x^2} \quad \frac{\rho R}{\rho a} = \tan \alpha$$

$$V = \frac{1}{2} \int_0^a (a^2 - x^2) \tan \alpha dx$$

旋转体体积

$$V_x = \pi \int_a^b f(x)^2 dx$$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 围成平面图形绕 x 轴旋转一周所得球体积

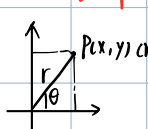
$$V = \int_0^a \pi \frac{b^2}{a^2} (a^2 - x^2) dx = 2\pi \frac{b^2}{a^2} \left(ax - \frac{x^3}{3} \right) \Big|_0^a = \frac{8}{3} \pi a b^2$$


微元法

- ① 分割 $[X_{i-1}, X_i] (i=1 \dots n)$
 - ② 近似 (求积微元) $\Delta V_i \approx f(x_i) \Delta x_i$
 - ③ 求和
 - ④ 极限 $V = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x_i = \int_a^b f(x) dx$
- ① 取近似乘微元 $\Delta V \approx f(x) \Delta x$ 舍去高阶无穷小
- ② 得微分 $dV = f(x) dx$
- ③ $V = \int_a^b f(x) dx$

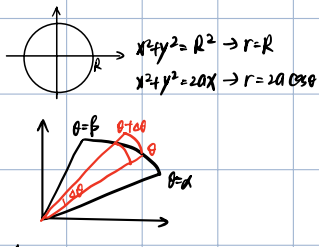
微元法及应用

2. 曲边扇形 (平面图形 S 在极坐标中算)

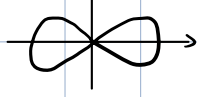


$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \quad \begin{cases} 0 \leq r < +\infty \\ 0 \leq \theta < 2\pi \end{cases}$$

$$\begin{cases} r = \sqrt{x^2 + y^2} \\ \theta = \arctan \frac{y}{x} \end{cases}$$

$$S_0 = \frac{1}{2} \int_0^\theta r^2 d\theta = \frac{1}{2} \int_0^\theta r^2(\theta) d\theta$$


算 $(x^2 + y^2)^2 = x^2 - y^2$ 所围 S

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \quad \begin{cases} r^4 = r^2 \cos^2 \theta - r^2 \sin^2 \theta \\ \Rightarrow r^2 = \cos 2\theta \end{cases}$$


起 A: $x=1, y=0, r=1, \theta=0$

终 B: $x=0, y=0, r=0, \theta=\frac{\pi}{4}$

$$S = 4 \cdot \frac{1}{2} \int_0^{\frac{\pi}{4}} r^2 d\theta = 1$$

3. 弧长计算公式

定义 曲线 Γ 上任意取点 $A = M_0, M_1, \dots, M_{n-1}, M_n = B$

$$S_n = \sum_{i=1}^n |M_{i-1} M_i| \quad \lim_{n \rightarrow \infty} S_n \text{ 称为弧长 } (\Gamma \text{ 可求长})$$

曲线 Γ 参数方程 $\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases}$ 其中 $t \in [a, b]$ 上函数连续且 $\varphi'(t) + \psi'(t) \neq 0$, 则 Γ 为光滑曲线

$$\Delta s \approx \sqrt{[\varphi(t+\Delta t) - \varphi(t)]^2 + [\psi(t+\Delta t) - \psi(t)]^2}$$

$$= \sqrt{[\varphi'(\xi) \Delta t]^2 + [\psi'(\eta) \Delta t]^2} \quad (t \leq \xi, \eta \leq t + \Delta t)$$

$$= \sqrt{\varphi'^2(\xi) + \psi'^2(\eta)} \Delta t \approx \sqrt{\varphi'^2(t) + \psi'^2(t)} \Delta t$$

当 $\Delta t \rightarrow 0, \xi \rightarrow t, \eta \rightarrow t$

$$\lim_{\Delta t \rightarrow 0} \frac{\sqrt{\varphi'^2(\xi) + \psi'^2(\eta)} \Delta t}{\Delta t} = \sqrt{\varphi'^2(t) + \psi'^2(t)}$$

$$ds = \sqrt{\varphi'^2 + \psi'^2} dt$$

$$s = \int_a^b \sqrt{\varphi'^2 + \psi'^2} dt \quad \text{把 } \begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases} \text{ 换 } \begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

$$s = \int_a^b \sqrt{1 + f'^2(x)} dx = \int_c^d \sqrt{1 + \psi'^2(y)} dy = \int_\alpha^\beta \sqrt{r'^2(\theta) + r_{\theta}^2} d\theta$$

4. 弧微分公式

$$s(t) = \int_{t_0}^t \sqrt{\varphi'^2 + \psi'^2} dt \quad (a \leq t \leq b)$$

$$\frac{ds}{dt} = \sqrt{\varphi'^2 + \psi'^2} > 0 \Rightarrow ds = \sqrt{\varphi'^2 + \psi'^2} dt$$

$$\therefore ds = \sqrt{1 + f'^2(x)} dx = \sqrt{1 + \psi'^2(y)} dy = \sqrt{r'^2(\theta) + r_{\theta}^2} d\theta = \sqrt{(\rho x')^2 + (\rho y')^2}$$

N 何意 $|dp| = \sqrt{(dx)^2 + (dy)^2}$

5. 旋转体体积及侧面积



考虑 $[x, x+dx]$ 上 S 微元 $S_{微元} = 2\pi \frac{y+\sqrt{y^2+z^2}}{2} \sqrt{1+y'^2} dx = 2\pi y \sqrt{1+y'^2} dx + \pi \frac{z}{y} \sqrt{1+y'^2} dx$

$V_x = \int_a^b \pi y^2 dx$ $V_y = 2\pi \int_a^b x |f(x)| dx$

求星形线周长 $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}} (a>0)$

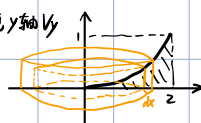
① $\frac{2}{3}x^{-\frac{1}{3}} + \frac{2}{3}y^{-\frac{1}{3}} \cdot y' = 0 \Rightarrow y = \pm \sqrt{\frac{y}{x}}$

$S = 4 \int_0^a \sqrt{1+y'^2} dx = 4 \int_0^a \left(\frac{a}{x}\right)^{\frac{1}{3}} dx = \dots = 6a$

② $S = 4 \int_0^{\frac{\pi}{2}} \sqrt{(a \cos t)^2 + (a \sin t)^2} dt = 4 \int_0^{\frac{\pi}{2}} a dt = 2\pi a$

D为 $4y=x^2$ 与 $x=2$ 及 x 轴围 (D绕 x 轴 V_x 绕 y 轴 V_y)

壳层法 $dV = 2\pi x \cdot y \cdot dx$
 $V_x = \int_0^2 2\pi x \cdot \frac{x^2}{4} dx$



算半径为 R 球面面积

① $S = 2\pi \int_{-R}^R y \sqrt{1+y'^2} dx = 2\pi \int_{-R}^R \sqrt{R^2-x^2} \cdot \frac{R}{\sqrt{R^2-x^2}} dx = 2\pi R^2$

$\begin{cases} x = R \cos t \\ y = R \sin t \end{cases}$

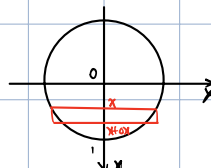
② $S = 2\pi \int_0^\pi y \sqrt{x'^2+y'^2} dt = 2\pi \int_0^\pi R \sin t \sqrt{(R \sin t)^2 + (R \cos t)^2} dt = 4\pi R^2$

1. 液体静压力 (对壁)

取 $[x, x+dx]$ $\Delta F = (\rho g x) \cdot (\rho y \cdot dx)$

$dF = 2\rho g x \sqrt{1+x'^2} dx$

$F = \int_0^1 2\rho g x \sqrt{1+x^2} dx = \frac{3\rho g}{5}$



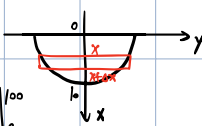
某一流体
对各个方向压强相等

2. 变力做功

$\Delta V = \pi y^2 \Delta x$ $\Delta W = \rho g \pi (100-x^2) x \Delta x$

$dW = \rho g \pi (100-x^2) x dx$

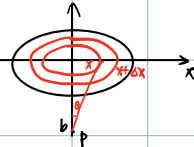
$W = \int_0^{10} \rho g \pi (100-x^2) x dx = \frac{1}{2} \rho g \pi (100-x^2) \Big|_0^{10} = 2500 \rho g \pi$



3. 引力

取 $[x, x+dx]$ $\Delta m = \rho \pi x \Delta x$

$\Delta F_y = \frac{G \Delta m m}{b^2+x^2} \cos \theta = 2G \rho \pi \frac{b x}{(b^2+x^2)^{\frac{3}{2}}} \Delta x$

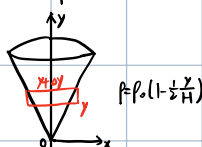


4. 质量

取 $[y, y+dy]$ $\Delta M = \rho \cdot (1-\frac{y}{H}) \cdot \pi (\frac{b}{H} y)^2 \cdot dy$

$dM = \frac{\rho \pi b^2}{H^3} (1-\frac{y}{H}) y^2 dy$

$M = \frac{\rho \pi b^2}{H^3} \int_0^H y^2 (1-\frac{y}{H}) dy = \frac{1}{24} \pi \rho b^2 H$



$\rho \cdot (1-\frac{y}{H})$

5. 动能 $E = \frac{1}{2} m v^2 = \frac{1}{2} m r^2 \omega^2$

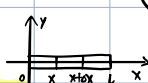
取 $[r, r+dr]$ $\Delta m = \rho \cdot 2\pi r dr$

$\Delta E = \frac{1}{2} (\Delta m) v^2 = \pi \rho r dr r^2 \omega^2$



6. 转动惯量 $J = m x^2$

$\Delta J = \rho \Delta x x^2$



7. 函数平均值 $\bar{y} = \frac{1}{b-a} \int_a^b f(x) dx$ 积分中值定理

取 $[X_{i-1}, X_i]$ 右端点 X_i

$\bar{y} = \lim_{n \rightarrow \infty} \frac{f(x_1) + \dots + f(x_n)}{n} = \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{f(x_i)}{n} = \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{f(x_i)}{X_i - X_{i-1}} \Delta x_i = \frac{1}{b-a} \int_a^b f(x) dx$

$\bar{v} = \frac{1}{T} \int_0^T (u + g t) dt = \frac{1}{2} (u + v)$

$\bar{i} = \frac{1}{2\pi} \int_0^{2\pi} I_m \sin \omega t dt = \frac{1}{\sqrt{2}} I_m$

定义 $f(x)$ 在 $[a, +\infty)$ 上连续, $\forall t > a$, $\int_a^t f(x) dx$ 存在

$\int_a^{+\infty} f(x) dx \stackrel{\text{def}}{=} \lim_{t \rightarrow +\infty} \int_a^t f(x) dx$ 为 $f(x)$ 在 $[a, +\infty)$ 上的第一类反常积分 / 无穷限的广义积分

① $\int_a^{+\infty} f(x) dx = \lim_{t \rightarrow +\infty} \int_a^t f(x) dx$

转化为变上限积分 $x \rightarrow \infty$ 的极限

② $\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx$

③ $\int_{-\infty}^{+\infty} f(x) dx = \int_{-\infty}^a f(x) dx + \int_a^{+\infty} f(x) dx$ 且 $\int_{-\infty}^{+\infty} f(x) dx$ 收敛 $\Leftrightarrow \int_{-\infty}^a f(x) dx$ 与 $\int_a^{+\infty} f(x) dx$ 均收敛

极限: 发散+发散=发散/收敛 共用一个 $f(x)$ 符号
积分: 发散+发散=发散

易错: $\int_{-\infty}^{+\infty} x dx = \int_{-\infty}^0 x dx + \int_0^{+\infty} x dx$ 发散
 $\int_{-\infty}^0 x dx = \frac{1}{2} x^2 \Big|_{-\infty}^0 = -\lim_{x \rightarrow -\infty} \frac{1}{2} x^2$ 不收敛
正确: $\int_{-\infty}^{+\infty} x dx = -\lim_{x \rightarrow -\infty} \frac{1}{2} x^2 + \lim_{x \rightarrow +\infty} \frac{1}{2} x^2$

$\lim_{t \rightarrow +\infty} \left(\int_{-t}^t x dx + \int_t^{+\infty} x dx \right)$

只有当两个 $f(x)$ 均收敛才能合并

定积分才能
正负面积抵消

讨论无穷限积分的敛散性

(1) $\int_0^{+\infty} x e^{-x} dx = -\int_0^{+\infty} x d e^{-x} = -x e^{-x} - \int_0^{+\infty} -e^{-x} dx = -\lim_{x \rightarrow +\infty} x e^{-x} + e^{-x} \Big|_0^{+\infty} = 0 - (-1) = 1$

(2) $\int_e^{+\infty} \frac{1}{x \ln x} dx = \int_e^{+\infty} \frac{1}{\ln x} d \ln x = \lim_{x \rightarrow +\infty} \ln |\ln x| \Big|_e^{+\infty}$ 发散

(3) $\int_{-\infty}^{+\infty} \sin x dx = \int_{-\infty}^0 \sin x dx + \int_0^{+\infty} \sin x dx$ 发散

讨论 $\int_a^{+\infty} \frac{dx}{x^p} (a>0)$ 敛散性 第一类反常积分

① $p > 1$ $\int_a^{+\infty} \frac{dx}{x^p} = \lim_{t \rightarrow +\infty} \int_a^t \frac{dx}{x^p} = \lim_{t \rightarrow +\infty} \left[\frac{x^{-p+1}}{-p+1} \right]_a^t = \lim_{t \rightarrow +\infty} \frac{1}{1-p} (t^{1-p} - a^{1-p}) = \begin{cases} \frac{a^{1-p}}{p-1} & p > 1 \\ +\infty & p < 1 \end{cases}$

② $p = 1$ $\int_a^{+\infty} \frac{dx}{x} = \lim_{t \rightarrow +\infty} \ln x \Big|_a^t = +\infty$

Tip

$\int_a^{+\infty} \frac{dx}{x^p} = \begin{cases} \text{收敛} & p > 1 \\ \text{发散} & p \leq 1 \end{cases}$

定积分在物理中应用

无穷区间

反常积分

原函数: $\int_a^b f(x)dx = \lim_{n \rightarrow \infty} [F(x_1) - F(x_0)] \stackrel{\text{def}}{=} F(b) - F(a) = F(x) \Big|_a^b$

$\int_{-\infty}^b f(x)dx = \lim_{a \rightarrow -\infty} [F(b) - F(a)] \stackrel{\text{def}}{=} F(b) - F(-\infty) = F(x) \Big|_{-\infty}^b$

$\int_a^{\infty} f(x)dx \stackrel{\text{def}}{=} F(\infty) - F(a) = F(x) \Big|_a^{\infty}$

$\int_0^{\infty} \frac{dx}{(x+1)^2 \sqrt{x^2+1}}$

① 例变换 $\frac{t-1}{t^2-1} \int_0^1 \frac{t^2}{t^2-1} dt = \int_0^1 \frac{t^2}{t^2-1} dt = \int_0^1 \frac{t^2}{t^2-1} dt = -\frac{1}{2} \ln|t^2-1| + \frac{1}{2} \int_0^1 \frac{1}{t^2-1} dt$

② 三角换元 $x = \sec \theta$

$\int_1^{\infty} \frac{\arctan x}{x^2} dx$ 不同类型函数积分 分部 1次x1次 拆分

$= -\int_1^{\infty} \arctan x d\frac{1}{x} = -\frac{\arctan x}{x} \Big|_1^{\infty} + \int_1^{\infty} \frac{dx}{x(1+x^2)}$

$= -\frac{\pi}{4} + \frac{1}{2} \int_1^{\infty} \frac{dx}{x^2(1+x^2)} = -\frac{\pi}{4} + \frac{1}{2} \int_1^{\infty} (\frac{1}{x^2} - \frac{1}{1+x^2}) dx$ Tip

定义: $f(x)$ 在 $[a, b]$ 上连续, $\lim_{x \rightarrow a^+} f(x) = a$, 则 a 为瑕点

$\int_a^b f(x)dx$ 为无界函数 $f(x)$ 在 $[a, b]$ 上的第二类反常积分 转化为变上限积分 $x \rightarrow 0$ 的极限

① a 为瑕点, 则 $\int_a^b f(x)dx = \lim_{\epsilon \rightarrow 0^+} \int_{a+\epsilon}^b f(x)dx$ $\forall c \in (a, b), \int_a^c f(x)dx$ 为 ϵ 的函数

② b 为瑕点, 则 $\int_a^b f(x)dx = \lim_{\epsilon \rightarrow 0^+} \int_a^{b-\epsilon} f(x)dx$

③ 设内点 $c \in (a, b)$ 为瑕点, 则 $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$ 且 $\int_a^b f(x)dx$ 收敛 $\Leftrightarrow \int_a^c f(x)dx$ 与 $\int_c^b f(x)dx$ 均收敛

讨论 $\int_a^b \frac{dx}{(b-x)^p}$ ($b > a$) 敛散性 第2类反常积分

$x=b$ 为瑕点

① $p \neq 1$ $\int_a^b \frac{dx}{(b-x)^p} = \lim_{\epsilon \rightarrow 0^+} \int_a^{b-\epsilon} \frac{dx}{(b-x)^p} = \lim_{\epsilon \rightarrow 0^+} \frac{-1}{p-1} (b-x)^{1-p} \Big|_a^{b-\epsilon} = \lim_{\epsilon \rightarrow 0^+} \frac{(b-a)^{1-p} - \epsilon^{1-p}}{1-p} = \begin{cases} \frac{(b-a)^{1-p}}{1-p} & p < 1 \\ +\infty & p > 1 \end{cases}$ Tip

② $p=1$ $\int_a^b \frac{dx}{(b-x)^p} = \lim_{\epsilon \rightarrow 0^+} \int_a^{b-\epsilon} \frac{1}{b-x} dx = -\lim_{\epsilon \rightarrow 0^+} \ln|b-x| \Big|_a^{b-\epsilon} = \lim_{\epsilon \rightarrow 0^+} [\ln(b-\epsilon) - \ln(b-a)] = +\infty$

Tip $\int_a^b \frac{dx}{(b-x)^p} \begin{cases} p < 1 \text{ 收敛} \\ p \geq 1 \text{ 发散} \end{cases}$

原函数: $\int_a^b f(x)dx = F(x) \Big|_a^b = F(b) - F(a)$

a 瑕点 $F(x) \stackrel{\text{def}}{=} \lim_{x \rightarrow a^+} F(x)$ b 瑕点 $F(x) \stackrel{\text{def}}{=} \lim_{x \rightarrow b^-} F(x)$

$\int_{\frac{1}{2}}^1 \frac{dx}{\sqrt{1-x^2}}$ $x=1$ 瑕点 $\sqrt{1-x^2}$

$= \int_{\frac{1}{2}}^1 \frac{dx}{\sqrt{1-x^2}} = \int_{\frac{1}{2}}^1 \frac{dx}{\sqrt{(\frac{1}{2})^2 - (x-\frac{1}{2})^2}} = \arcsin(2x-1) \Big|_{\frac{1}{2}}^1 = \frac{\pi}{2}$

$+ \int_1^{\frac{3}{2}} \frac{dx}{\sqrt{x^2-x}} = \int_1^{\frac{3}{2}} \frac{dx}{\sqrt{(x-\frac{1}{2})^2 - (\frac{1}{2})^2}} = \ln \left[(x-\frac{1}{2}) + \sqrt{(x-\frac{1}{2})^2 - (\frac{1}{2})^2} \right] \Big|_1^{\frac{3}{2}} = \ln(1+\sqrt{2})$

无界函数

定理1(比较判别法)

设 $f(x), g(x)$ 在 $[a, +\infty)$ 上连续, 且有 $0 \leq f(x) \leq g(x)$

(1) 若 $\int_a^{\infty} g(x)dx$ 收敛, 则 $\int_a^{\infty} f(x)dx$ 收敛

(2) 若 $\int_a^{\infty} f(x)dx$ 发散, 则 $\int_a^{\infty} g(x)dx$ 发散

(1) $\exists M > 0, \exists t \in [a, +\infty) \int_a^t f(x)dx \leq \int_a^t g(x)dx \leq M$

(2) 反证 若收敛则加收

推论1 设 $f(x)$ 在 $[a, +\infty)$ 上连续

(1) 若 x 充分大时, 有 $0 \leq f(x) \leq \frac{c}{x^p}$ ($p > 1, c > 0$) 收敛

(2) 若 x 充分大时, 有 $f(x) \geq \frac{c}{x^p}$ ($p \leq 1, c > 0$) 发散

推论2(极限形式)

设 $f(x)$ 在 $[a, +\infty)$ ($a > 0$) 上连续, 且 $\lim_{x \rightarrow \infty} \frac{f(x)}{x^p} = A$

(i) 若 $0 < A < +\infty$ $f(x) \sim \frac{A}{x^p}$ ($x \rightarrow +\infty$)

则 $\int_a^{\infty} f(x)dx$ 与 $\int_a^{\infty} \frac{A}{x^p} dx$ 敛散性相同 $\begin{cases} p > 1 \int_a^{\infty} f(x)dx \text{ 收敛} \\ p \leq 1 \int_a^{\infty} f(x)dx \text{ 发散} \end{cases}$

(ii) 若 $A=0$, $f(x) \geq 0$ 且 $p \geq 1$, 则 $\int_a^{\infty} f(x)dx$ 收敛

(iii) 若 $A=+\infty$ 且 $p \leq 1$, 则 $\int_a^{\infty} f(x)dx$ 发散

判敛散性

(1) $\int_1^{\infty} \frac{dx}{x^p \sqrt{x^2+1}}$ $\frac{1}{x^p \sqrt{x^2+1}} \sim \frac{1}{x^{p+\frac{1}{2}}} \sim \frac{1}{x^{\frac{5}{2}}} (x \rightarrow +\infty)$ $A=1, p=\frac{5}{2} > 1 \therefore$ 收敛

(2) $\int_1^{\infty} \frac{\arctan x}{x^2} dx$ $\frac{\arctan x}{x^2} \sim \frac{\pi}{x^2}$ $A=\frac{\pi}{2}, p=2 > 1 \therefore$ 收敛

(3) 极限积分 $\int_a^{\infty} e^{-x} dx$ 由 $\lim_{x \rightarrow \infty} \frac{x^p}{e^x} = 0$ ($a > 0$) $\lim_{x \rightarrow \infty} \frac{x^2}{e^x} \stackrel{\text{令 } x^2=t}{=} \lim_{t \rightarrow \infty} \frac{t}{e^t} = 0$ $A=0, p=2 > 1 \therefore$ 收敛

对 $f(x)$ 化为 $-f(x)$

对可变量函数不可判 $\rightarrow \int_a^{\infty} |f(x)|dx$

定理2(绝对收敛准则)

设 $f(x)$ 在 $[a, +\infty)$ 上连续, 若 $\int_a^{\infty} |f(x)|dx$ 收敛, 则 $\int_a^{\infty} f(x)dx$ 收敛

$\because 0 \leq f(x) + |f(x)| \leq 2|f(x)| \therefore \int_a^{\infty} [f(x) + |f(x)|]dx$ 收敛

$\therefore \int_a^{\infty} f(x)dx = \int_a^{\infty} [f(x) + |f(x)|]dx - \int_a^{\infty} |f(x)|dx$ 收敛

(1) 当 $\int_a^{\infty} |f(x)|dx$ 收敛, 称 $\int_a^{\infty} f(x)dx$ 绝对收敛

先判 $\int_a^{\infty} |f(x)|dx$

性质: (1) $\int_a^{\infty} f(x)dx = \int_a^c f(x)dx + \int_c^{\infty} f(x)dx$

(2) $\int_a^{\infty} k f(x)dx = k \int_a^{\infty} f(x)dx$

(3) $\int_a^{\infty} [f(x) \pm g(x)]dx = \int_a^{\infty} f(x)dx \pm \int_a^{\infty} g(x)dx$

(4) $\int_a^{\infty} u dv = uv \Big|_a^{\infty} - \int_a^{\infty} v du$

(5) 换元

(ii) $\lim_{x \rightarrow \infty} \frac{f(x)}{x^p} = A$, 对 $\epsilon = \frac{A}{2} > 0, \exists b > 0$, 当 $x > b$ 时, $-\frac{A}{2} < \frac{f(x)}{x^p} - A < \frac{A}{2} \Rightarrow 0 < \frac{A}{2} < \frac{f(x)}{x^p} < \frac{3A}{2} \frac{1}{x^p}$

(iii) $\lim_{x \rightarrow \infty} \frac{f(x)}{x^p} = 0$, 对 $\epsilon = 1, \exists b > 0$, 当 $x > b$ 时, $|\frac{f(x)}{x^p}| < 1$ 而 $f(x) \geq 0 \Rightarrow 0 < f(x) < \frac{1}{x^p}$

(iii) $\lim_{x \rightarrow \infty} \frac{f(x)}{x^p} = +\infty$, 对 $M=1, \exists b > 0$, 当 $x > b$ 时, $\frac{f(x)}{x^p} > 1$

由推论1

敛散性判别法

无穷区间

② 若 $\int_a^{+\infty} f(x) dx$ 发, 而 $\int_a^{+\infty} |f(x)| dx$ 收, 称 $\int_a^{+\infty} f(x) dx$ 条件收敛

例 $\int_a^{+\infty} \frac{\sin x^2}{x^2} dx$ 敛散性

$\because |\frac{\sin x^2}{x^2}| \leq \frac{1}{x^2}$ $\int_a^{+\infty} \frac{1}{x^2} dx$ 收 $\therefore \int_a^{+\infty} |\frac{\sin x^2}{x^2}| dx$ 收 $\therefore$ 绝对收敛

定理: $f(x), g(x)$ 在 $[a, b]$ 上连续, 且 x 充分靠近 b 时, 有 $0 \leq f(x) \leq g(x)$

(1) 若 $\int_a^b f(x) dx$ 收, 则 $\int_a^b g(x) dx$ 收

(2) 若 $\int_a^b f(x) dx$ 发, 则 $\int_a^b g(x) dx$ 发

推论1: $f(x)$ 在 $[a, b]$ 上连续, 当 x 充分靠近 b 时, 有 $0 \leq f(x) \leq \frac{c}{(b-x)^p}$ ($c > 0, p < 1$) 则 $\int_a^b f(x) dx$ 收敛
若 $f(x) \geq \frac{c}{(b-x)^p}$ 则 $\int_a^b f(x) dx$ 发散

推论2: 加在 $[a, b]$ 上连续, 若 $\lim_{x \rightarrow b} \frac{f(x)}{(b-x)^p} = \lim_{x \rightarrow b} (b-x)^p f(x) = A$

(i) 若 $0 < A < +\infty$, $f(x) \sim \frac{A}{(b-x)^p}$ ($x \rightarrow b^-$)

$\int_a^b f(x) dx$ 与 $\int_a^b \frac{A}{(b-x)^p} dx$ 敛散性相同 $\begin{cases} p < 1 & \text{收} \\ p \geq 1 & \text{发} \end{cases}$

(ii) 若 $A = 0$, 且 x 充分靠近 b 时, $f(x) \geq 0$ 且 $p < 1$ 时, $\int_a^b f(x) dx$ 收

(iii) 若 $A = +\infty$ 且 $p > 1$ 时, $\int_a^b f(x) dx$ 发

定理: $f(x)$ 在 $[a, b]$ 上连续, b 为瑕点

(1) 若 $\int_a^b |f(x)| dx$ 收, 称 $\int_a^b f(x) dx$ 绝对收敛

(2) 若 $\int_a^b |f(x)| dx$ 发, 而 $\int_a^b f(x) dx$ 收, 称 $\int_a^b f(x) dx$ 条件收敛

例 敛散性

(1) $\int_1^{+\infty} \frac{x^{p+1}}{\sqrt{x^2-1}} dx$ $f(x) \geq 0$ 且当 $x \rightarrow 1^+$ 时 $f(x) \sim \frac{x^{p+1}}{\sqrt{(x-1)(x^2+x+1)}} \sim \frac{\sqrt{x}}{(x-1)^{\frac{1}{2}}}$ $p = \frac{1}{2} < 1$ 收

(2) $\int_1^{+\infty} \frac{x^{p+1}}{\sqrt{x^2-1}} dx$ 由 (1) $\int_1^{+\infty} \frac{x^{p+1}}{\sqrt{x^2-1}} dx$ 收

$\int_1^{+\infty} \frac{x^{p+1}}{\sqrt{x^2-1}} dx$ 在 $[2, +\infty)$ 上连续且 $f(x) \geq 0$, $f(x) \sim \frac{x^{p+1}}{\sqrt{x^2-1}} \sim \frac{x^p}{x^{\frac{1}{2}}} = x^{p-\frac{1}{2}}$ $p < 1$ 也

定义: 讨论 $\int_0^{+\infty} x^{s-1} e^{-x} dx$ 敛散性 $\Rightarrow s > 0$ 收, 否则发 $\Gamma(\text{Gamma})$ 函数 $\Gamma(s) = \int_0^{+\infty} x^{s-1} e^{-x} dx$ ($s > 0$)

$\int_0^{+\infty} x^{s-1} e^{-x} dx = \int_0^1 x^{s-1} e^{-x} dx + \int_1^{+\infty} x^{s-1} e^{-x} dx = I_1 + I_2$ $\int_1^{+\infty} \square dx$ $\square$ 为瑕点

I_1 为第一类反常积分, $x=0$ 为瑕点

$x^{s-1} e^{-x} \sim \frac{1}{x^{1-s}} (x \rightarrow 0^+)$ $1-s < 1 \Rightarrow s > 0$ 收

I_2 为第一类反常积分, 由 $\lim_{x \rightarrow +\infty} \frac{x^s}{e^x} = 0$ ($s > 1, k$ 常) 推论2

$\lim_{x \rightarrow +\infty} x^s e^{-x} = \lim_{x \rightarrow +\infty} \frac{x^s}{e^x} = 0$ x 为任何实数均收敛

性质: $\Gamma(s+1) = s\Gamma(s)$ ($s > 0$)

$\Gamma(s+1) = \int_0^{+\infty} x^{s+1-1} e^{-x} dx = \int_0^{+\infty} x^s e^{-x} dx$

$= -\int_0^{+\infty} x^s d(e^{-x}) = -[x^s e^{-x}]_0^{+\infty} - \int_0^{+\infty} (-x^s) e^{-x} dx = s\Gamma(s)$

$\Rightarrow \Gamma(n+1) = n\Gamma(n) = \dots = n!\Gamma(1) = n!$ $\Gamma(1) = \int_0^{+\infty} e^{-x} dx = 1$

定义域的延拓

$\Gamma(s) = \frac{\Gamma(s+1)}{s}$ $s < 0 \Rightarrow s+1 > 0$

$= \frac{\Gamma(s+2)}{s(s+1)}$ $s < -1 \Rightarrow s+2 > 0$ 定义域 $s > -2$ ($s=0, -1$ 除外)

类推: 在除去0与负整数外的 $\mathbb{R}$ 上有意义

计算 $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

$\therefore \int_0^{+\infty} e^{-x} dx = \sqrt{\frac{\pi}{2}}$

$\therefore \Gamma(\frac{1}{2}) = \int_0^{+\infty} x^{\frac{1}{2}-1} e^{-x} dx = \int_0^{+\infty} \frac{e^{-x}}{\sqrt{x}} dx = \int_0^{+\infty} e^{-x} dx \stackrel{t=\sqrt{x}}{=} \int_0^{+\infty} e^{-t^2} 2t dt = \sqrt{\pi}$

无界函数

Γ 函数